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Saving Energy on PTX with PFE Power Off

An HPE Juniper Networking Techpost

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2023-07-27

HPE

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1 Saving Energy on PTX with PFE Power Off

Ramdas Machat - 07/27/2023

Let's test the real power saving on PTX platforms achieved when shutting down used and unused Packet Forwarding Engine (PFE).

1.1 Introduction

Power efficiency is key in today's ever-growing Service providers and Cloud providers networks. Network industry improved the power and carbon footprint over the years through real innovations in silicon: manufacturing process, various components integration on the die and the package, monitoring and manageability.

PTX routers offer the granular control of individual PFE without impacting the other units present on the line card or the system. Operators can power on, power off and restart them at will, saving significant amount of energy.

In this article, we will detail:

- the multiple ways to power off PFEs manually and how to configure the system to take actions on PFE in case of error
- the power saving achieved turning off PFEs on an LC1201 in a PTX10016 with DR4 inserted and without affecting the traffic flowing on other ASICs of the card
- the power saving achieved turning off PFE on LC1201 in the three chassis types (PTX10004, PTX10008 and PTX10016) but also on fixed platforms like PTX10001-36MR, all of them without any optics inserted

Scope: we will not cover other Juniper platforms (MX/ACX/QFX/EX/…), only PTX.

1.2 PFE Power Off and Error Action

Packet Forward Engine are at the heart of our routing devices. Routers and Line Cards contains one or several of them. Multiple PFEs are interconnected via a switch fabric and fabric SerDes.

We can imagine multiple situations where you would like to turn off a PFE.

The first one is obvious: when PFEs are not in use. That could be the case if you use Pay As You Grow (PAYG) model with just a portion of the system/linecard capability. Or you didn't provisioned all ports yet and you strategicaly positioned your current ports to keep one or several PFE unused. Turning these unused PFEs off will reduce the system power consumption.

A second use-case would be to power off specific PFEs during periods of low traffic, usually at night. Traffic engineering rules will make sure no packets traverse these PFEs, and you can automate their power off to save energy for a couple of hours.

Let's take for example a PTX10001-36MR to illustrate these scenarios

- with optics connected and forwarding traffic on PFE0
- without any optic modules on PFE1
- with optics connected but no traffic on PFE2

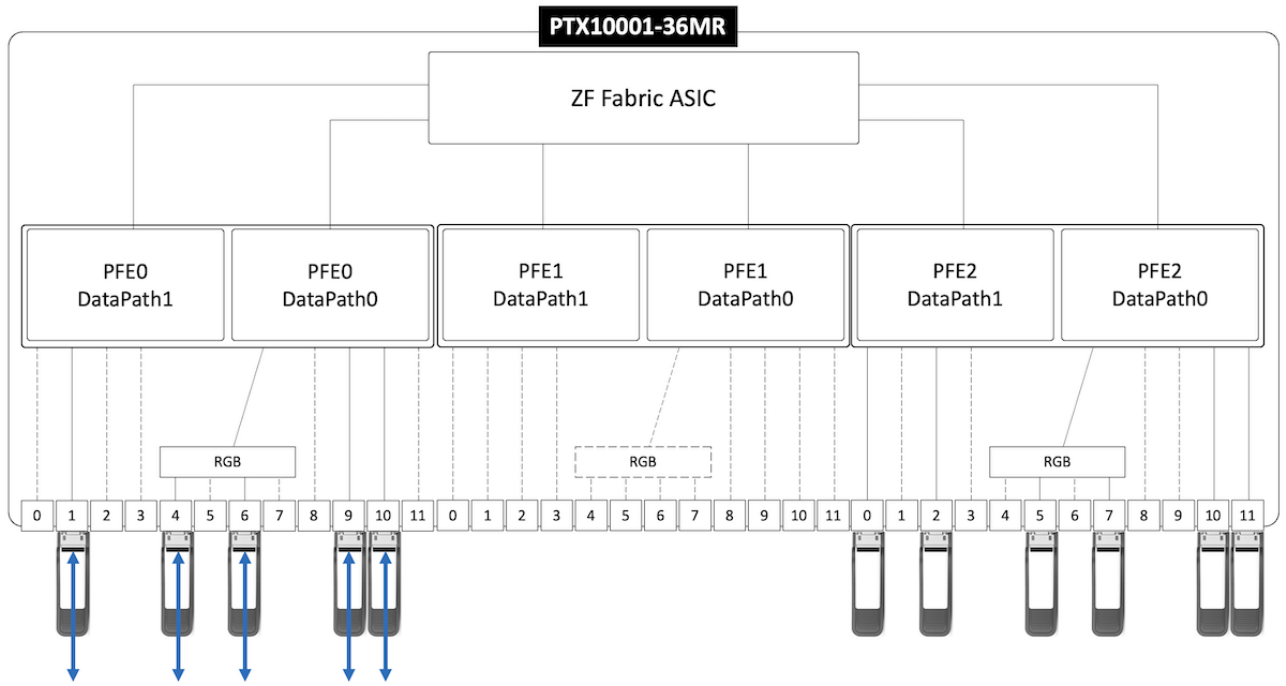


Figure 1: PTX10001-36MR Block Diagram

Several power saving features are activated by default on PTX:

- Empty ports SerDes are shutdown (example: et-0/0/0)
- Gearboxes with only empty ports are kept in reset mode, SerDes to the PFE and to the optic cages are shutdown (example: ports et-0/1/4 to et-0/1/7)

The network operator can decide to go further than the default behavior and shutdown / power off the PFE1, since no ports are using it. That's our first use-case described above.

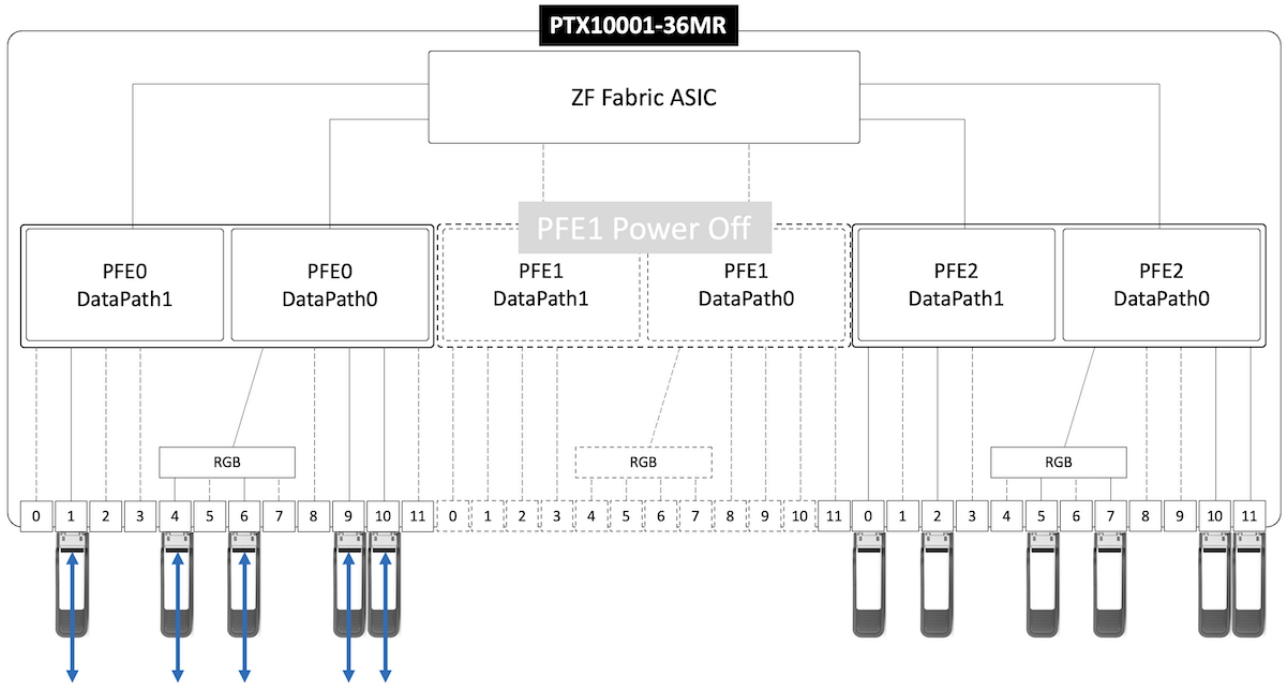


Figure 2: PTX10001-36MR with PFE1 Powered Off

That's the tests we will perform on various chassis and fixed platform in the second part of this article. It will give us a good idea of the saving we can get from the PFE only (since the ports, gearboxes and SerDes are shutdown already).

Some users could decide to go even further and power off PFE2 (with optics but no traffic):

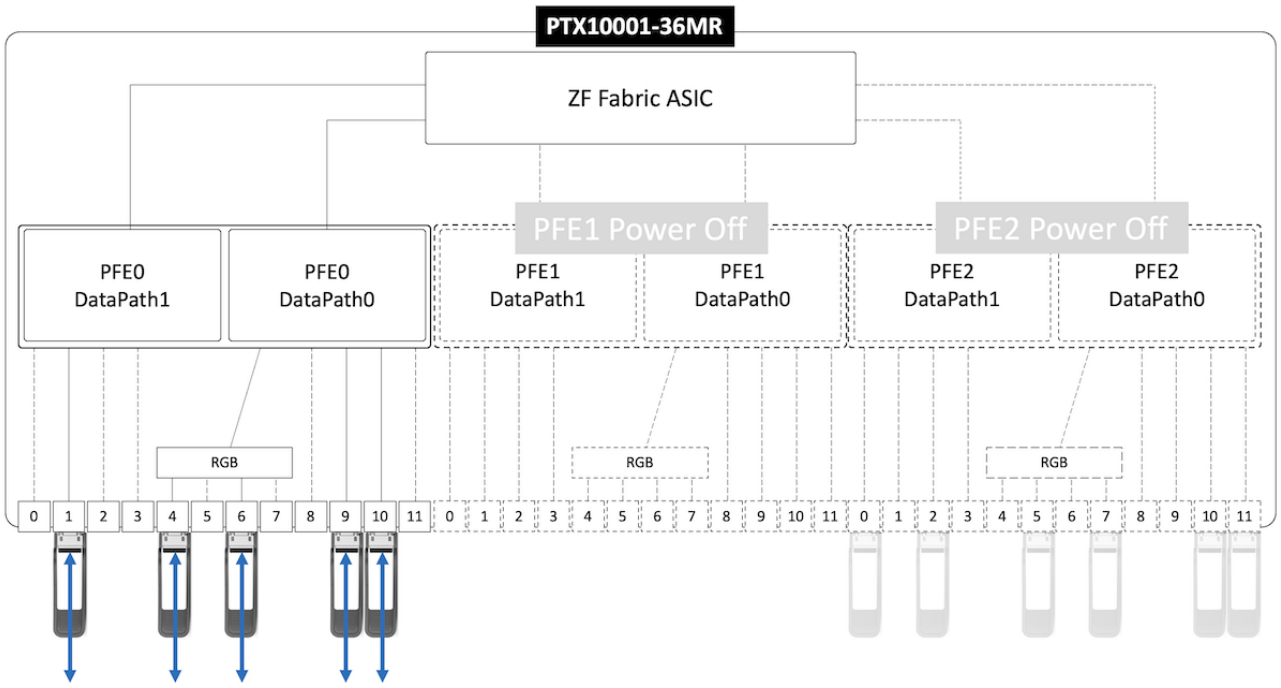


Figure 3: PTX10001-36MR with PFE1 and PFE2 Powered Off

Of course, all ports on PFE1 and PFE2 disappeared and will only be restored after we turn the PFEs on again.

These are the tests we will run on the first part of this techpost, specifically on a PTX10016 and LC1201. It will give us an idea of the savings achieved with optics, SerDes and PFE being turned off.

We picked the PTX10001-36MR for illustration of the use-cases but it's equally applicable for other PTX10003 and PTX10k chassis and line cards.

1.2.1 Supported Platforms

The handling of PFEs independently of the FPC (and other PFEs) is available in Junos EVO-only platforms and the Table1 below displays the first releases supporting this feature.

Table 1: Supported Hardware and Software release

Line Card	Number of PFEs	Platform	Supported Release
FPC-JNP10003	80C - 8160C - 16	PTX10003-80CPTX10003-160C	22.4R1-EVO
JNP10K-LC1201	5	PTX10008	21.2R1-EVO
JNP10K-LC1201	5	PTX10004/PTX10016	21.4R1-EVO

Line Card	Number of PFEs	Platform	Supported Release
JNP10K-LC1202	2	PTX10004/10008/10016	21.4R1-EVO
FPC-JNP10001-36MR	3	PTX10001-36MR	22.3R1-EVO

1.2.2 PFE States

Here are the possible PFE states which can be observed by running the CLI command “show chassis fpc pfe-instances all”

Table 2: PFE States

PFE State	Description
ONLINE	PFE is Online
Configured Power Off	PFE is Offline via Configuration on router
Offlined by CLI	PFE Offline via Request CLI operation by user on router
Offlined by CMError	PFE Offline via CMError Action
Transition_OFFLINE	PFE Transition status
FAULT	When “disable-pfe” CMError action is taken by the Router on Major/Fatal errors
PowerOn	PFE state during PFE power on transition
RESYNC	This is the state when the PFE is synchronizing the state with other components withing the system
READY	When the PFE is ready after power on but not online yet. Takes a few more seconds to bring up online

1.3 PFE Power Off

We will carry out tests on Express4-based PTX platforms: modular chassis (4, 8 and 16 slots) with 36x400GE LC1201 linecards, and on PTX10001-36MR fixed router. Routers were running Junos EVO versions: 22.4R2.11-EVO and 23.2R1.15-EVO

1.3.1 Configuration and CLI

PFE can be independently Powered off/on using Configuration mode or Operational (CLI) mode.

Below Table 3 shows the support of this feature across the platforms.

Table 3: Platforms Support for PFE Power On/Off

Mode	PTX10004	PTX10008	PTX10016	PTX10001-36MR	PTX10003-80C	PTX10003-160C
Config	sqrt	sqrt	sqrt	sqrt	sqrt	sqrt
CLI	sqrt	sqrt	sqrt	sqrt	X	X

Configuration Method

```
ptx10k16-01# set chassis fpc 0 pfe 0 power ?
Possible completions:
  off Power off PFE
  on Power on PFE

ptx10k16-01# set chassis fpc 0 pfe 0 power off
ptx10k16-01# commit
```

CLI Method

```
ptx10k16-01> request chassis fpc slot 0 pfe-instance 0 ?
Possible completions:
  no-forwarding No forwarding
  offline Take FPC or PFE-instance offline
  online Bring FPC or PFE-instance online
  restart Restart FPC or PFE-instance
  tag Firmware tag for online-reload

ptx10k16-01> request chassis fpc slot 0 pfe-instance 0 offline
Fru Block offline initiated!

ptx10k16-01>
```

The “operation” CLI methods doesn’t need to commit a configuration, which could be an advantage is certain cases.

Regardless the method, it takes less than 5s to transition to offline state.

1.4 Test1: PFE PowerOff/Offline with Optics and Traffic

In this first test, we will apply the two methods for two PFEs 0 and 1, and we will verify it does not impact other PFEs in the system.

We will use a LC1201 in a PTX10016 (16-slot chassis), offering 36 ports 400GE and based of 5x Express4. Each PFE package contains itself two datapaths.

All ports are populated with DR4 optics but only the last 20 are connected to a traffic generator pushing 50% full-meshed traffic, 512 bytes per packet.

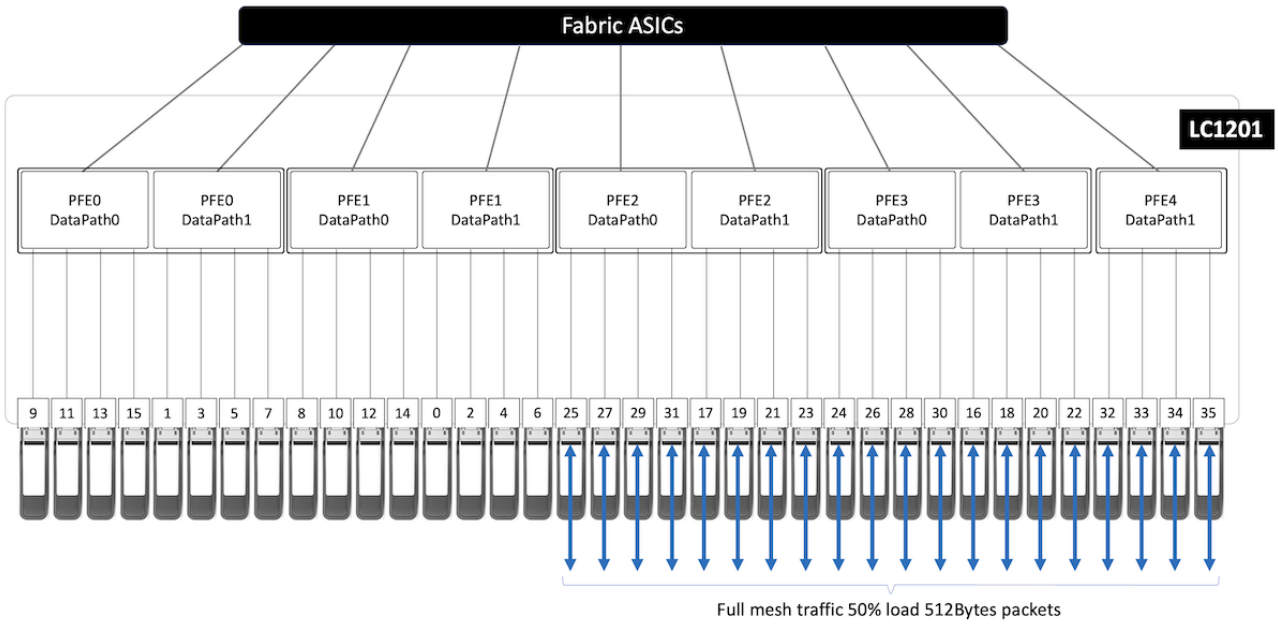


Figure 4: Test 1 Topology with LC1201, DR4 Optics and Traffic on PFE2/3/4

“‘ ptx10k16-01> show interfaces terse Interface Admin Link Proto Local Remote et-0/0/0 up up et-0/0/0.0 up up inet 100.1.1.1/24
multiservice pfh-0/0/0 up up pfh-0/0/0.16383 up up inet
et-0/0/2 up up et-0/0/2.0 up up inet 100.1.3.1/24
multiservice et-0/0/4 up up et-0/0/4.0 up up inet 100.1.5.1/24
multiservice et-0/0/6 up up et-0/0/6.0 up up inet 100.1.7.1/24
multiservice et-0/0/8 up up et-0/0/8.0 up up inet 100.1.9.1/24
multiservice et-0/0/10 up up et-0/0/10.0 up up inet 100.1.11.1/24
multiservice et-0/0/12 up up et-0/0/12.0 up up inet 100.1.13.1/24
multiservice et-0/0/14 up up et-0/0/14.0 up up inet 100.1.15.1/24
multiservice et-0/0/16 up up et-0/0/16.0 up up inet 100.1.17.1/24
multiservice et-0/0/17 up up et-0/0/17.0 up up inet 100.1.18.1/24
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multiservice et-0/0/34 up up et-0/0/34.0 up up inet 100.1.35.1/24
multiservice et-0/0/35 up up et-0/0/35.0 up up inet 100.1.36.1/24
multiservice

ptx10k16-01> show chassis fpc pfe-instance all
FPC 0 PFE-Instance PFE-State 0 ONLINE
1 ONLINE
2 ONLINE
3 ONLINE
4 ONLINE

ptx10k16-01> show chassis pic fpc-slot 0 pic-slot 0
FPC slot 0, PIC slot 0 information: Type JNP10K-36QDD-LC-PIC State Online
PIC version 00.00 Uptime 15 hours, 44 minutes, 25 seconds Port speed information:

Port PFE Capable Port Speeds 0 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 1 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 2 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 3 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 4 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 5 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 6 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 7 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 8 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 9 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 10 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 11 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 12 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 13 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 14 1 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 15 0 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 16 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 17 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 18 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 19 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 20 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 21 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 22 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 23 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 24 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 25 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 26 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 27 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 28 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 29 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 30 3 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 31 2 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 32 4 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 33 4 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 34 4 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G 35 4 1x10G 4x10G 1x40G 4x25G 1x100G 2x50G 2x100G 3x100G 4x100G 1x400G

ptx10k16-01>



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